

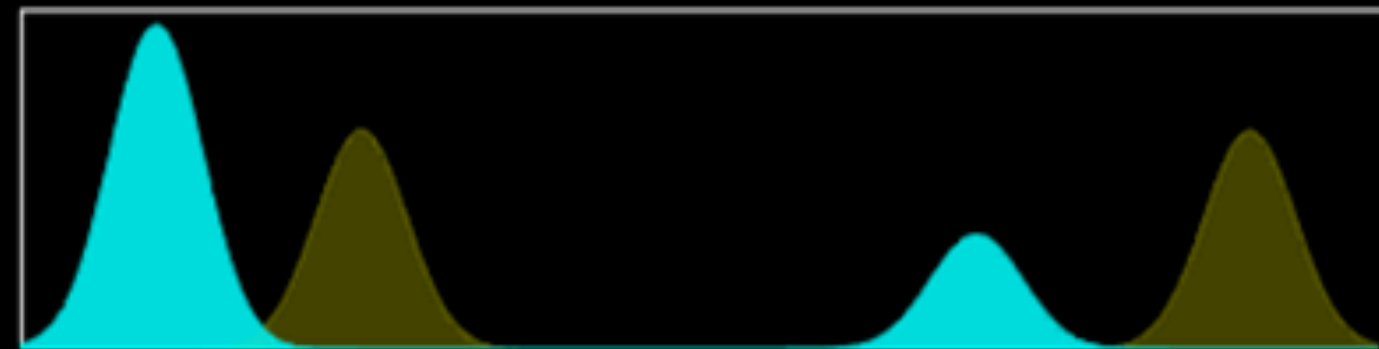
necessary if $\sum_i \mathbf{a}_i \neq \sum_j \mathbf{b}_j$

can help regularize

Unbalanced OT: Intuition

- Unbalanced OT is **necessary** if $\sum_i \mathbf{a}_i \neq \sum_j \mathbf{b}_j$ (exact OT is not defined)
- Even if $\sum_i \mathbf{a}_i = \sum_j \mathbf{b}_j = 1$, unbalanced OT **can help regularize**.
- Mass can appear or disappear with cost $\rightarrow$ smoother interpolations.
- E.g., if outliers are expensive to match, paying $\tau_1 D_\psi(\pi \mathbf{1}_m \mid \mathbf{a})$ might be better than paying matching cost.

[Un]-balanced Barycenters



Classical OT ($\tau = +\infty$)